Midterm Review

DS-6030 | Spring 2026

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1 Lectures

1.1 Supervised Learning Foundations

1.1.1 Core Concepts

- Supervised learning framework: Learn a function from labeled data that minimizes expected loss on future data.
- Training vs test error
- Prediction vs inference
- Loss functions (RMSE, MAE, log-loss)
- Bias–variance tradeoff
- Expected Prediction Error (EPE)
- Overfitting vs underfitting
- Model flexibility and complexity; effective degrees of freedom (edf)
- Prediction Pipeline
- Confidence Intervals

1.1.2 Resampling

- Bootstrap and OOB
- K-fold and Monte Carlo cross-validation
- Repeated cross-validation
- Using CV for tuning
- Using CV for performance estimation
- When a final test set is or is not necessary
- Nested cross-validation (structure and purpose)

1.1.3 Tuning (or hyper) parameters

- What is a tuning parameter?
- Grid search
- Resampling based tuning; mathematical correction to training data (e.g., AIC/BIC)
- Separation of tuning vs evaluation data
- Optimism of performance from tuning on same data

1.2 Elastic Net / Penalized Regression

- Lasso (L1)
- Ridge (L2)
- Elastic net mixing parameter
- Shrinkage and regularization
- Effect of penalty on bias and variance

1.3 Classification

1.3.1 Binary Classification Framework

- Hard labels vs probabilities
- Decision thresholds; know how to calculate
- Confusion matrix
- False positives / false negatives
- Cost-sensitive classification
- logistic regression

1.3.2 Performance Metrics

- assessing probability estimates vs. hard labels
- Log Loss, Brier Score
- Accuracy / misclassification
- Sensitivity / specificity
- ROC curves
- AUC and its drawbacks
- Relationship between ranking and probability quality

1.4 Tree-Based Methods

1.4.1 Probability Trees

- Recursive partitioning
- Splitting criteria
- Controlling tree complexity

1.4.2 Random Forests

- Bagging
- Feature subsampling (mtry, num_features)
- Variance reduction
- Tree decorrelation
- Number of trees and stability
- Bias–variance in forests

1.4.3 Out-of-Bag (OOB)

- Bootstrap samples leave out ~37% of observations
- OOB prediction per observation
- OOB error estimation
- OOB vs cross-validation
- Using OOB for tuning

1.5 Other Models

1.5.1 SVM

- Hinge Loss
- Comparison with logistic regression
- Tuning Parameters: Cost (or penalty), Kernel choice, Kernel specific
- kernel formulation
- dealing with unbalanced outcomes; dealing with non-equal misclassfication costs

1.5.2 Generative Classifiers

- Bayes Theorem
- Generative vs. Discriminative Models
  - P(Y|X) [Discriminative]
  - P(Y,X) [Generative]
- LDA/QDA

1.6 Predictive Models

- Form of prediction
- Loss function used to set model parameters
- Tuning parameters and how they related to bias and variance
- Model Families:
  - Linear Regression
  - Logistic Regression
  - Nearest Neighbor
  - Trees
  - Random Forests / Bagging
  - SVM
  - LDA/QDA

1.7 Risk Score Calibration

- Note: On the Take Home Exam; not tested in class
- Converting Risk Score to calibrated probability
- Approaches:
  - logistic regression
  - splines
  - binning
  - isotonic splines

2 Homework

2.1 HW 1

- Optimal Performance (theoretical MSE)
- True (quadratic) model structure may not be optimal predictor
- Effect of training and test data size on model selection
- Effect of training and test data size on predictive performance

2.2 HW 2

- Predictive Modeling Pipeline
- Assessing Predictive Performance including confidence intervals
- Single Train/Validation
- Cross-Validation
- Nested Cross-Validation

2.3 HW 3

- Random Forest Tuning
- Confidence Interval on predictive performance
- Tree splitting

2.4 HW 4

- Linear Regression (penalized) for binary outcomes
- Logistic Regression (penalized) for binary outcomes
- Random Forest Tuning
- ROC Curves: based on training data vs. test data
- Resampling based ROC curves
- Predicting Probability
- Predicting Label
- log loss
- Expected misclassification cost

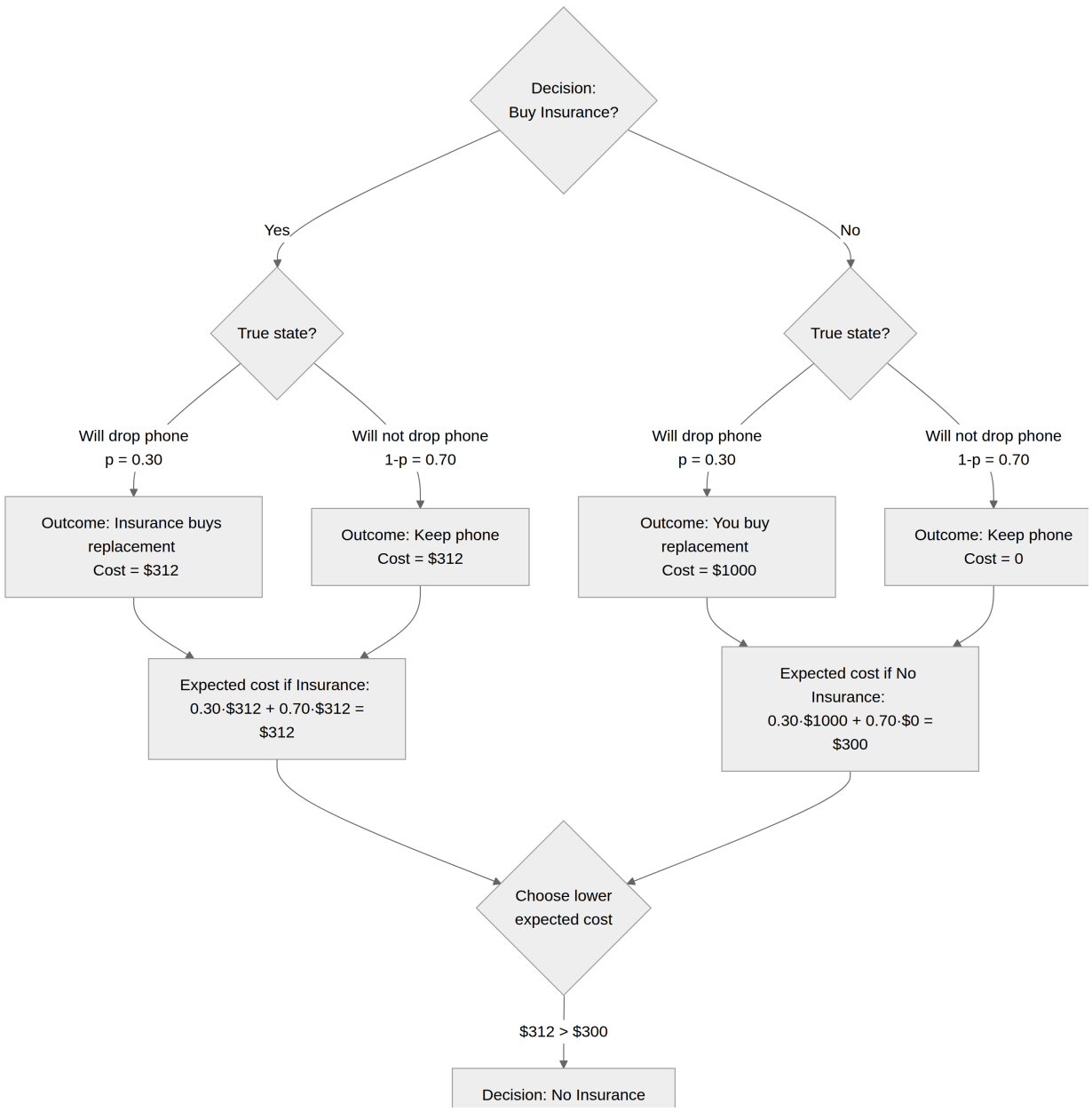
3 Review

3.1 Binary Classification Evaluation

3.1.1 Phone Insurance Problem

Should you buy insurance on your new phone?

- Probability you will drop and break your phone: $p = 0.30$
- Cost of buying replacement phone (no insurance): \$1000
- Cost of insurance paid phone: \$412
  - Cost of insurance over two years: \$312 (24 months at \$13/month)
  - Cost of deductible: \$100



3.1.2 Optimal Threshold for Crime Linkage

- Same Offender: $Y = 1$
- Different Offenders: $Y = 0$
- Costs: $C_{\text{FP}} = 1$, $C_{\text{FN}} = 8$

Case	Outcome	Cost
FP	$Y = 0,\ \hat{Y} = 1$	$C_{\text{FP}} = 1$
TP	$Y = 1,\ \hat{Y} = 1$	0
FN	$Y = 1,\ \hat{Y} = 0$	$C_{\text{FN}} = 8$
TN	$Y = 0,\ \hat{Y} = 0$	0

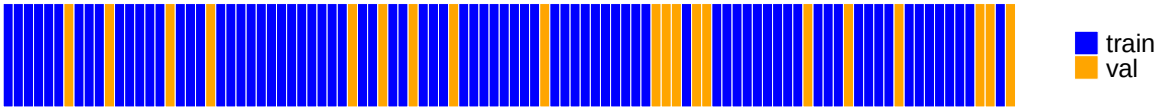
- Let $\hat{p} = \widehat{\Pr}(Y = 1 \mid X = x)$
- Should we classify as linked or unlinked?

3.2 Calculating Confidence Intervals

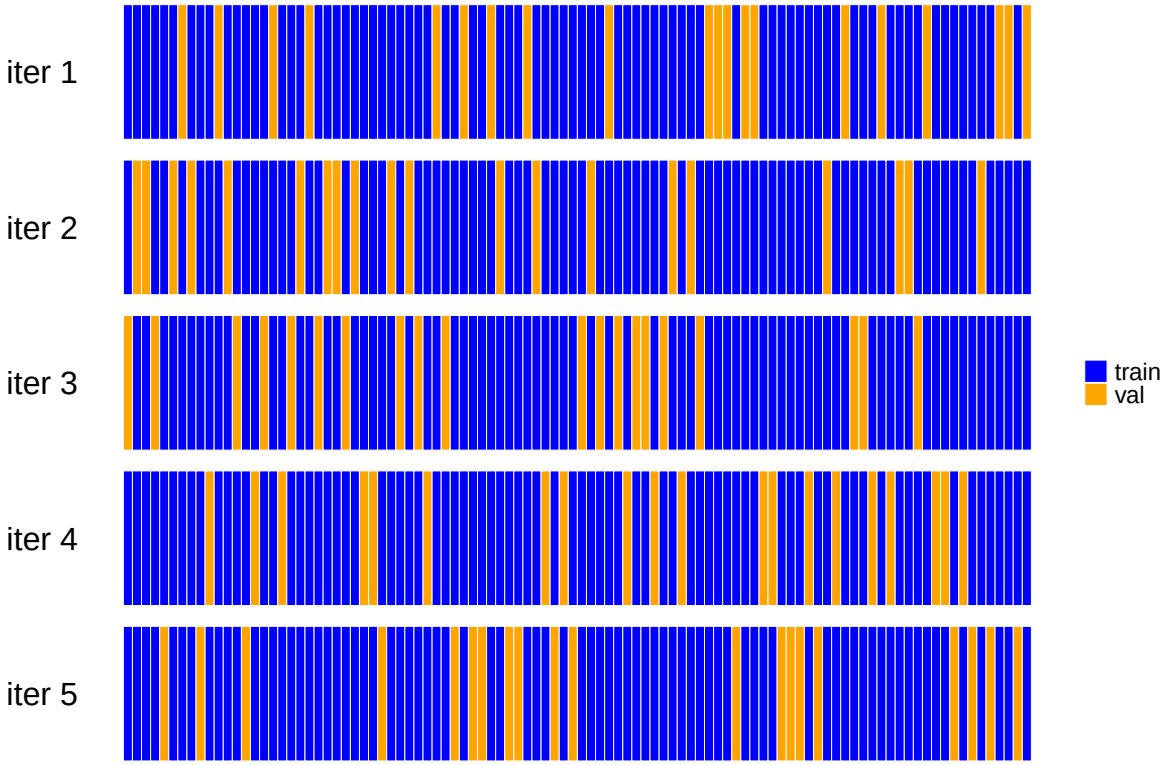
$$\begin{aligned}\text{CI} &= \text{Point Estimate} \pm \text{Margin of Error} \\ &= \text{Point Estimate} \pm (\text{Critical Value}) \cdot (\text{Standard Error})\end{aligned}$$

3.3 Estimating Model Performance

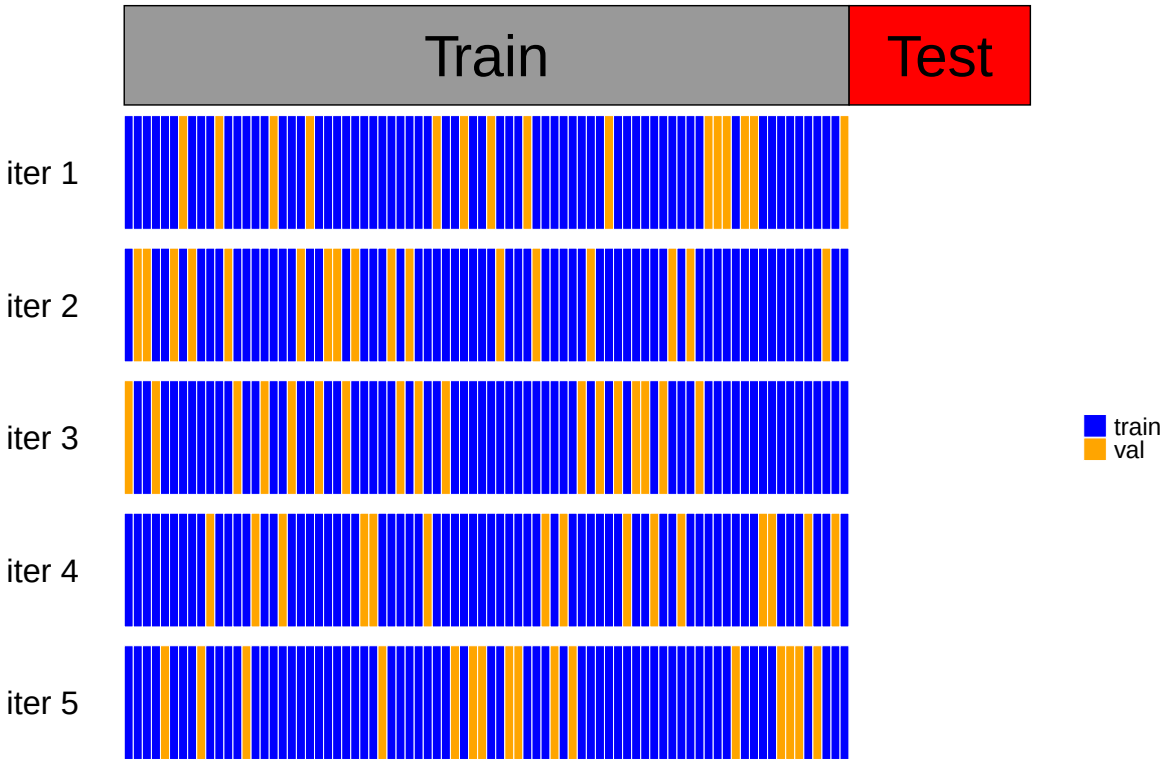
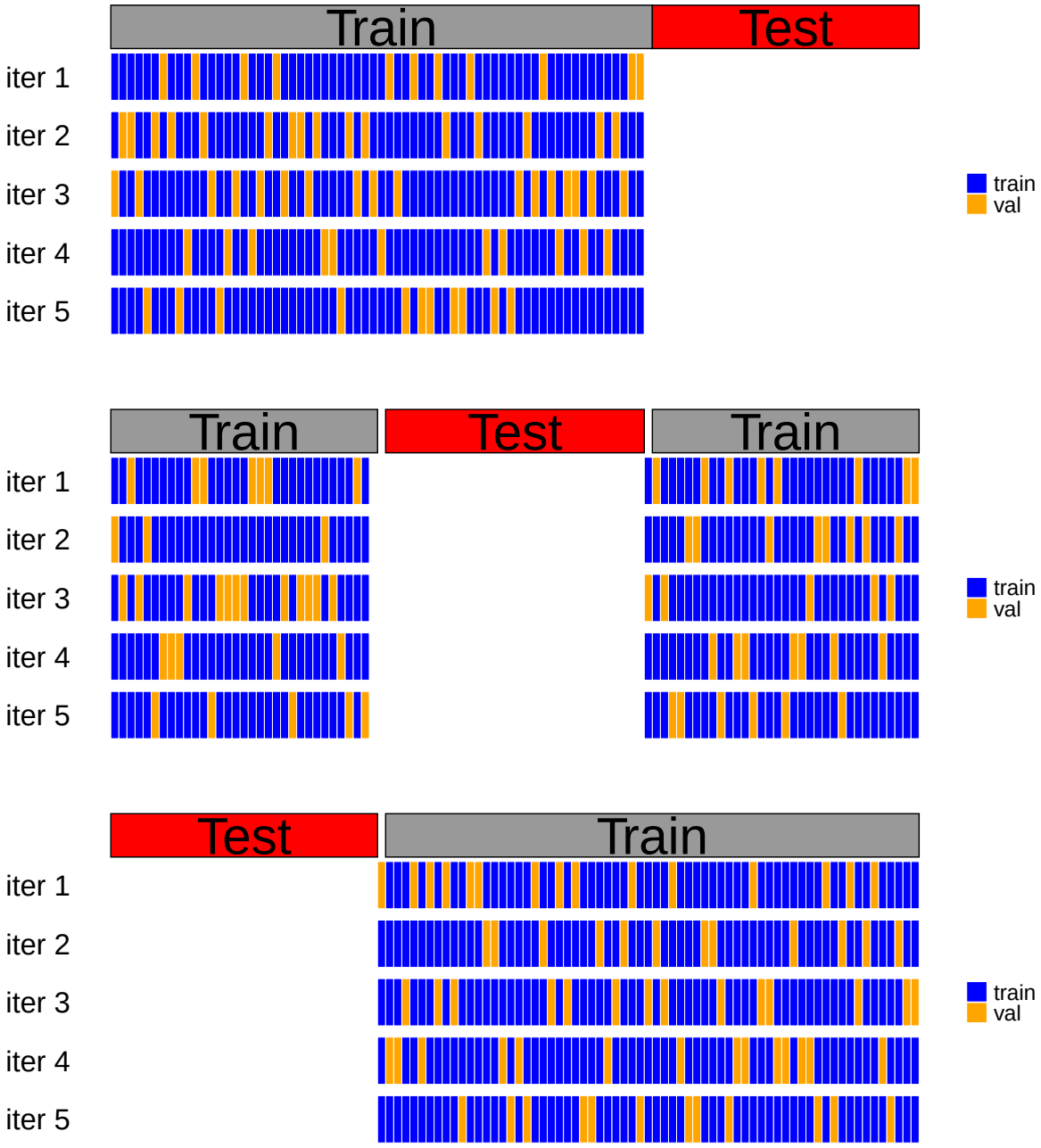
3.3.1 Single Train / Val Split



3.3.2 Cross-Validation (Multiple Train/Val splits)



3.3.3 Cross-Valiation + Final Test Set



3.3.4 Nested Cross-Validation